EXP.NO: 9	Develop an Android Calculator Application using Android Studio
DATE: 10.03.2026	

AIM:

To develop an Android application that implements a basic calculator using Android controls such as Button, TextView, and EditText. The calculator will support basic mathematical operations like Addition, Subtraction, Multiplication, and Division.

ALGORITHM:

- 1. Initialize the Android Application:
- o Set up a new Android project in Android Studio.
- o Create a MainActivity Java or Kotlin file for the main screen of the application.

- 2. Design the User Interface (UI):
- o Create the following UI components in the XML layout file:

EditText for input fields where the user will enter numbers (two EditTexts for number1 and number2).

Buttons for each arithmetic operation (Addition, Subtraction, Multiplication, Division).

TextView to display the result of the operation.

3. Declare Variables:

o Create variables for the EditText fields, Buttons, and TextView.

o Define variables to hold the input numbers and the result (e.g., num1, num2, result).

4. Get Input from the User:

o Get the values entered by the user in EditText fields using getText().toString() and convert them to numerical values (like Integer.parseInt() or Double.parseDouble()).

5. Set Event Listeners for Buttons:

o For each button (Addition, Subtraction, Multiplication, and Division), set an OnClickListener.

o Inside the onClick() method for each button, perform the respective arithmetic operation:

Addition: result = num1 + num2

Subtraction: result = num1 - num2

Multiplication: result = num1 * num2

Division: result = num1 / num2 (add validation to prevent division by zero).

6. Perform Arithmetic Operation:

o Check which button was clicked (using view.getId()), and based on the button, perform the respective arithmetic operation.

o Handle Division by Zero: Check if num2 is 0 before performing division and show an appropriate error message if true.

7. Display the Result:

o Use TextView to display the result of the operation.

o Convert the result to a string and set it using setText() method on the TextView.

8. Validate User Inputs:

o Before performing any operation, validate the input to ensure the user has

entered valid numbers (not empty or non-numeric characters).

9. Handle Edge Cases:

o Ensure the app handles cases where users input invalid values or attempt division by zero. Display appropriate messages for errors.

10. Finish the Operation:

o Once the result is displayed, the user can either perform another calculation or exit the app..

CODE:

activity_main.xml

```
&lt;?xml version=&quot;1.0&quot; encoding=&quot;utf-8&quot;?&gt;
&lt;androidx.constraintlayout.widget.ConstraintLayout
xmlns:android=&quot;http://schemas.android.com/apk/res/android&quot;
xmlns:app=&quot;http://schemas.android.com/apk/res-auto&quot;
xmlns:tools=&quot;http://schemas.android.com/tools&quot;
android:layout_width=&quot;match_parent&quot;
android:layout_height=&quot;match_parent&quot;
tools:context=&quot;.MainActivity&quot;&gt;
&lt;!-- First Number Input --&gt;
&lt;EditText
android:id=&quot;@+id/editTextNumber1&quot;
android:layout_width=&quot;0dp&quot;
android:layout_height=&quot;wrap_content&quot;
android:inputType=&quot;numberDecimal&quot;
android:hint=&quot;Enter first number&quot;
android:textSize=&quot;18sp&quot;
android:textColorHint=&quot;#000000&quot;
android:typeface=&quot;monospace&quot;
```

```
android:textStyle=&quot;bold&quot;
app:layout_constraintTop_toTopOf=&quot;parent&quot;
app:layout_constraintStart_toStartOf=&quot;parent&quot;
app:layout_constraintEnd_toEndOf=&quot;parent&quot;
android:layout_margin=&quot;16dp&quot; /&gt;
&lt;!-- Second Number Input --&gt;
&lt;EditText
android:id=&quot;@+id/editTextNumber2&quot;
android:layout_width=&quot;0dp&quot;
android:layout_height=&quot;wrap_content&quot;
android:inputType=&quot;numberDecimal&quot;
android:hint=&quot;Enter second number&quot;
android:textSize=&quot;18sp&quot;
android:textColorHint=&quot;#000000&quot;
android:typeface=&quot;monospace&quot;
android:textStyle=&quot;bold&quot;
app:layout_constraintTop_toBottomOf=&quot;@id/editTextNumber1&quot;
app:layout_constraintStart_toStartOf=&quot;parent&quot;
app:layout_constraintEnd_toEndOf=&quot;parent&quot;
android:layout_margin=&quot;16dp&quot; /&gt;
&lt;!-- Row of Operation Buttons --&gt;
&lt;LinearLayout
android:id=&quot;@+id/buttonRow&quot;
android:layout_width=&quot;0dp&quot;
android:layout_height=&quot;wrap_content&quot;
android:orientation=&quot;horizontal&quot;
android:weightSum=&quot;4&quot;
android:layout_margin=&quot;16dp&quot;
app:layout_constraintTop_toBottomOf=&quot;@id/editTextNumber2&quot;
app:layout_constraintStart_toStartOf=&quot;parent&quot;
app:layout_constraintEnd_toEndOf=&quot;parent&quot;&gt;
```

```
&lt;Button
android:id=&quot;@+id/buttonAdd&quot;
android:layout_width=&quot;0dp&quot;
android:layout_height=&quot;wrap_content&quot;
android:layout_weight=&quot;1&quot;
android:text=&quot;Add&quot; /&gt;
&lt;Button
android:id=&quot;@+id/buttonSubtract&quot;
android:layout_width=&quot;0dp&quot;
android:layout_height=&quot;wrap_content&quot;
android:layout_weight=&quot;1&quot;
android:text=&quot;Subtract&quot; /&gt;
&lt;Button
android:id=&quot;@+id/buttonMultiply&quot;
android:layout_width=&quot;0dp&quot;
android:layout_height=&quot;wrap_content&quot;
android:layout_weight=&quot;1&quot;
android:text=&quot;Multiply&quot; /&gt;
&lt;Button
android:id=&quot;@+id/buttonDivide&quot;
android:layout_width=&quot;0dp&quot;
android:layout_height=&quot;wrap_content&quot;
android:layout_weight=&quot;1&quot;
android:text=&quot;Divide&quot; /&gt;
&lt;/LinearLayout&gt;
&lt;!-- Clear Button --&gt;
&lt;Button
android:id=&quot;@+id/buttonClear&quot;
android:layout_width=&quot;wrap_content&quot;
android:layout_height=&quot;wrap_content&quot;
android:text=&quot;Clear&quot;
```

```
app:layout_constraintTop_toBottomOf=&quot;@id/buttonRow&quot;
app:layout_constraintStart_toStartOf=&quot;parent&quot;
app:layout_constraintEnd_toEndOf=&quot;parent&quot;
android:layout_marginTop=&quot;12dp&quot;/&gt;
&lt;!-- Result TextView --&gt;
&lt;TextView
android:id=&quot;@+id/textViewResult&quot;
android:layout_width=&quot;wrap_content&quot;
android:layout_height=&quot;wrap_content&quot;
android:text=&quot;Result will appear here&quot;
android:textSize=&quot;18sp&quot;
android:textStyle=&quot;bold&quot;
app:layout_constraintTop_toBottomOf=&quot;@id/buttonClear&quot;
app:layout_constraintStart_toStartOf=&quot;parent&quot;
app:layout_constraintEnd_toEndOf=&quot;parent&quot;
android:layout_marginTop=&quot;24dp&quot; /&gt;
&lt;/androidx.constraintlayout.widget.ConstraintLayout&gt;
```

MainActivity.java*{

```
package com.example.calculatorapp;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
public class MainActivity extends AppCompatActivity {
EditText editTextNumber1, editTextNumber2;
Button buttonAdd, buttonSubtract, buttonMultiply, buttonDivide, buttonClear;
TextView textViewResult;
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

editTextNumber1 = findViewById(R.id.editTextNumber1);
editTextNumber2 = findViewById(R.id.editTextNumber2);
buttonAdd = findViewById(R.id.buttonAdd);
buttonSubtract = findViewById(R.id.buttonSubtract);
buttonMultiply = findViewById(R.id.buttonMultiply);
buttonDivide = findViewById(R.id.buttonDivide);
buttonClear = findViewById(R.id.buttonClear);
textViewResult = findViewById(R.id.textViewResult);

buttonAdd.setOnClickListener(view -&gt; calculate(&#39;+&#39;));

buttonSubtract.setOnClickListener(view -&gt; calculate(&#39;-&#39;));

buttonMultiply.setOnClickListener(view -&gt; calculate(&#39;*&#39;));

buttonDivide.setOnClickListener(view -&gt; calculate(&#39;/&#39;));

buttonClear.setOnClickListener(view -&gt; {
editTextNumber1.setText(&quot;&quot;);
editTextNumber2.setText(&quot;&quot;);
textViewResult.setText(&quot;Result will appear here&quot;);
});
}
private void calculate(char operator) {
```

```
String input1 = editTextNumber1.getText().toString();
String input2 = editTextNumber2.getText().toString();
if (input1.isEmpty() || input2.isEmpty()) {
textViewResult.setText(&quot;Please enter both numbers.&quot;);
return;
}
try {
double num1 = Double.parseDouble(input1);
double num2 = Double.parseDouble(input2);
double result = 0;
switch (operator) {
case '+':
result = num1 + num2;
break;
case '-':
result = num1 - num2;
break;
case '*':
result = num1 * num2;
break;
case '/':
if (num2 == 0) {
textViewResult.setText(&quot;Cannot divide by zero&quot;);
return;
}
result = num1 / num2;
break;
```

```
}
textViewResult.setText(&quot;Result: &quot; + result);
} catch (NumberFormatException e) {
textViewResult.setText(&quot;Invalid input&quot;);
}
}
}
```

OUTPUT:

Result:

An calculator is designed using Android studio which uses controls like Button, TextView, EditText having basic functionality like Addition, Subtraction, multiplication, and Division.